REPORT

Measuring m_π and m_{PCAC} in the two-flavor Schwinger model

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We show how to compute the analogue of the pion mass and the PCAC mass in the two-flavor Schwinger model with degenerate fermions. At the end of the report, we show some results that corroborate that our implementation of the HMC is indeed correct.

Some important conventions are

$$\gamma_0 = \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \gamma_1 = \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \gamma_5 = i\sigma_1\sigma_2 = \begin{pmatrix} -1 & 0 \\ 0 & 1. \end{pmatrix} \tag{1}$$

We simulate Wilson fermions, *i.e.* the Dirac operator is given by

$$\begin{aligned} D_{\alpha\beta}[\mathbf{n},\mathbf{m}] &= (m_0+2)\delta_{\alpha\beta}\delta_{\mathbf{n},\mathbf{m}} \\ &\quad - \frac{1}{2}\sum_{\mu=0}^{1}\left[(1-\sigma_\mu)_{\alpha\beta}\, U_\mu(\mathbf{n})\delta_{\mathbf{n}+\hat{\mu},\mathbf{m}} + (1+\sigma_\mu)_{\alpha\beta}\, U_\mu^\dagger(\mathbf{n}-\hat{\mu})\delta_{\mathbf{n}-\hat{\mu},\mathbf{m}}\right], \end{aligned} \tag{2}$$

where $\alpha, \beta \in \{0, 1\}$ represent the spin indices; $\mathbf{n}, \mathbf{m} \in \Lambda$; $U_\mu(\mathbf{n}) \in \mathrm{U}(1)$; m_0 is the bare mass parameter and σ_μ are the Pauli matrices. $\mu = 0$ refers to time and $\mu = 1$ to space. The bare mass is related to the so-called *hopping parameter* via

$$\kappa = \frac{1}{2(m_0+2)}. \tag{3}$$

In the literature results are commonly expressed in terms of this parameter[1]. Much of the things done in this report are based on Ref. [1].

1 Measuring the pion mass

In order to extract m_π we need to compute a two-point function that allows us to extract the mass from an exponential decay

$$\langle \mathcal{O}_\pi(t)\overline{\mathcal{O}}_\pi(0)\rangle = Ae^{-tm_\pi}. \tag{4}$$

We take

$$\mathcal{O}_\pi(n) = \overline{d}(n)\gamma_5 u(n) \tag{5}$$

[1]Note that the Dirac operator should be rescaled accordingly and that in QCD $\kappa = 1/(2(m_0+4))$. We are also fixing the Wilson parameter $r = 1$.

with n on the lattice. Let us calculate

$$\langle \mathcal{O}_\pi(n)\overline{\mathcal{O}}_\pi(m)\rangle \tag{6}$$

for two arbitrary lattice sites.

$$\begin{aligned}
\langle \mathcal{O}_\pi(n)\overline{\mathcal{O}}_\pi(m)\rangle &= \langle \overline{d}(n)\gamma_5 u(n)\overline{u}(m)\gamma_5 d(m)\rangle \\
&= (\gamma_5)_{\alpha\beta}(\gamma_5)_{\rho\sigma}\langle \overline{d}_\alpha(n)u_\beta(n)\overline{u}_\rho(m)d_\sigma(m)\rangle \\
&= -(\gamma_5)_{\alpha\beta}(\gamma_5)_{\rho\sigma}\langle u_\beta(n)\overline{u}_\rho(m)d_\sigma(m)\overline{d}_\alpha(n)\rangle \\
&= -(\gamma_5)_{\alpha\beta}(\gamma_5)_{\rho\sigma}\langle u_\beta(n)\overline{u}_\rho(m)\rangle\langle d_\sigma(m)\overline{d}_\alpha(n)\rangle \\
&= -(\gamma_5)_{\alpha\beta}(\gamma_5)_{\rho\sigma}D^{-1}_{\beta\rho}(n|m)D^{-1}_{\sigma\alpha}(m|n).
\end{aligned} \tag{7}$$

We used the anticommutativity of the spinors to rearrange them together with the Wick's theorem to contract the fields. Note that we can only contract fields of the same flavor. We also use the fact that both flavors have the same mass, so the propagator is the same

$$D^{-1}_{\alpha\beta}(n|m) = \langle u_\alpha(n)\overline{u}_\beta(m)\rangle = \langle d_\alpha(n)\overline{d}_\beta(m)\rangle. \tag{8}$$

Let us remember that

$$\gamma_5 D\gamma_5 = D^\dagger \iff \gamma_5 D^{-1}\gamma_5 = D^{-1\dagger} \iff (\gamma_5)_{\rho\sigma}D^{-1}_{\sigma\alpha}(m|n)(\gamma_5)_{\alpha\beta} = (D^{-1}_{\beta\rho}(n|m))^*. \tag{9}$$

Therefore

$$\langle \mathcal{O}_\pi(n)\overline{\mathcal{O}}_\pi(m)\rangle = -\sum_{\alpha\beta}|D^{-1}_{\alpha,\beta}(n|m)|^2 \tag{10}$$

If we project the first operator into momentum space we obtain

$$\langle \tilde{\mathcal{O}_\pi}(p,t)\overline{\mathcal{O}}_\pi(m_t,m_x)\rangle = \frac{1}{\sqrt{N_x}}\sum_x e^{in_x p}\langle \mathcal{O}_\pi(n_t,n_x)\overline{\mathcal{O}}_\pi(m_t,m_x)\rangle \propto e^{-n_t E(p)}, \tag{11}$$

with

$$E(p) \sim \sqrt{m_\pi^2 + p^2} \quad \text{(plus lattice artifacts)}. \tag{12}$$

To measure m_π we fix $p = 0$. For convenience, we set the source at zero as well. Thus, we are interested in measuring

$$C_\pi(t) = \frac{1}{\sqrt{N_x}}\sum_x \langle \mathcal{O}_\pi(t,x)\overline{\mathcal{O}}_\pi(0,0)\rangle = -\frac{1}{\sqrt{N_x}}\sum_x\sum_{\alpha\beta}|D^{-1}_{\alpha\beta}(t,x|0,0)|^2. \tag{13}$$

The normalization factor in front of the sum is irrelevant for extracting the pion mass. To compute the inverse matrix we can define two sources at the origin

$$\psi^{(0,\alpha_0)}_\alpha(m) = \delta_{m,0}\delta_{\alpha,\alpha_0}, \quad \alpha_0 = 0, 1. \tag{14}$$

and compute

$$D^{-1}(n|0)_{\alpha\alpha_0} = \sum_m\sum_\beta D^{-1}_{\alpha\beta}(n|m)\psi^{(0,\alpha_0)}_\beta(m). \tag{15}$$

This is equivalent to solving the Dirac equation using the source as a right-hand side. Essentially, we are extracting two columns of the inverse matrix. This can be done with a linear solver like BiCGStab, best case scenario.

The measurements of our correlator could be contaminated with excited states. However, we only want the lowest energy state. By measuring the *effective mass*

$$m_{\text{eff}}(t+1/2) = \ln \frac{C_\pi(t)}{C_\pi(t+1)} \tag{16}$$

one can observe when there is a plateau, which means that the correlator is dominated by the ground state energy and allows to fit $m_{\text{eff}} = m_\pi$. Since we are using periodic boundary conditions, we have

$$\frac{C_\pi(t)}{C_\pi(t+1)} = \frac{\cosh\left[m_{\text{eff}}(t - N_t/2)\right]}{\cosh\left[m_{\text{eff}}(t+1-N_t/2)\right]}. \tag{17}$$

In order to find the effective mass for each value of t, one has to solve this equation numerically. For instance, using the secant method.

2 Measuring the PCAC mass

To measure the renormalized fermion mass we use the Partially Conserved Axial Current Relation (PCAC), which in the continuum reads

$$\langle(\partial_\mu A^a_\mu(x))\mathcal{O}\rangle = 2m\langle P^a(x)\mathcal{O}\rangle, \tag{18}$$

where

$$A^a_\mu(x) = \overline{\psi}(x)\gamma_\mu\gamma_5\frac{\tau^a}{2}\psi(x), \quad P^a(x) = \overline{\psi}(x)\gamma_5\frac{\tau^a}{2}\psi(x) \tag{19}$$

are the axial current and pseudoscalar density. The operator $\mathcal{O}$ is arbitrary. The τ^a matrices act on flavor space. In our case, we have two flavors so we identify τ^a with the Pauli matrices. The PCAC relation remains the same after renormalization. Thus, we can use it to extract the renormalized mass from our simulations. On the lattice, the PCAC relation takes the form

$$\langle\left(\hat{\partial}_\mu A^a_\mu(n)\right)\mathcal{O}(n')\rangle = 2m\langle P^a(n)\mathcal{O}(n')\rangle \quad \text{(plus lattice artifacts).} \tag{20}$$

The symbol $\hat{\partial}$ indicates a discrete derivative. We use

$$\hat{\partial}_t f(t) = \frac{f(t+1) - f(t-1)}{2}. \tag{21}$$

To obtain a more explicit expression in terms of known quantities, let us choose $\mathcal{O} = P^a$ and analyze eq. (18)

$$\langle\left(\hat{\partial}_\mu A^a_\mu(n)\right)P^a(n')\rangle = \langle\hat{\partial}_0 A^a_0(n)P^a(n')\rangle + \langle\hat{\partial}_1 A^a_1(n)P^a(n')\rangle, \quad \text{(no Einstein summation on } a\text{).} \tag{22}$$

When we consider the projection into zero momentum space, essentially a sum over all the x variables, the second term will vanish, since it is a spatial derivative (a 1-divergence) and the boundaries are periodic. Therefore, we only consider the first term

$$\langle\left(\hat{\partial}_\mu A^a_\mu(n)\right)P^a(n')\rangle = \langle\hat{\partial}_0 A^a_0(n)P^a(n')\rangle = \frac{\langle A_0(t+1,x)P^a(t',x')\rangle - \langle A_0(t-1,x)P^a(t',x')\rangle}{2}. \tag{23}$$

Let us define fix the source at $(t', x') = (0, 0)$ from now on and define

$$C_{\text{AP}}(t) := \sum_x \langle A_0(t,x)P^a(0,0)\rangle. \tag{24}$$

In a similar manner we define

$$C_{\mathrm{PP}}(t) := \sum_x \langle P^a(t,x) P^a(0,0) \rangle. \tag{25}$$

Then

$$m(t) = \frac{C_{\mathrm{AP}}(t+1) - C_{\mathrm{AP}}(t-1)}{4 C_{\mathrm{PP}}(t)}. \tag{26}$$

We look for a plateau on this function, which we identify as the PCAC mass, m_{PCAC}.

We proceed to analyze each correlator independently. We choose $a = 3$ for convenience, although any other value should be equivalent. Thus, we replace

$$\tau^3 = \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{27}$$

For two flavors we have

$$\psi(n) = \begin{pmatrix} u(n) \\ d(n) \end{pmatrix}, \quad \overline{\psi}(n) = \left(\overline{u}(n), \overline{d}(n) \right). \tag{28}$$

Then

$$\begin{aligned}
\langle P^3(n) P^3(0) \rangle &= \frac{1}{4} \langle \overline{\psi}(n) \gamma_5 \sigma_3 \psi(n) \overline{\psi}(0) \gamma_5 \sigma_3 \psi(0) \rangle \\
&= \frac{1}{4} \langle \left(\overline{u}(n) \gamma_5 u(n) - \overline{d}(n) \gamma_5 d(n) \right) \left(\overline{u}(0) \gamma_5 u(0) - \overline{d}(0) \gamma_5 d(0) \right) \rangle \\
&= \frac{1}{4} \Big[\langle \overline{u}(n) \gamma_5 u(n) \overline{u}(0) \gamma_5 u(0) \rangle - \langle \overline{u}(n) \gamma_5 u(n) \overline{d}(0) \gamma_5 d(0) \rangle \\
&\quad - \langle \overline{d}(n) \gamma_5 d(n) \overline{u}(0) \gamma_5 u(0) \rangle + \langle \overline{d}(n) \gamma_5 d(n) \overline{d}(0) \gamma_5 d(0) \rangle \Big].
\end{aligned} \tag{29}$$

Let us analyze each term independently.

$$\begin{aligned}
\langle \overline{u}(n) \gamma_5 u(n) \overline{u}(0) \gamma_5 u(0) \rangle &= (\gamma_5)_{\alpha\beta} (\gamma_5)_{\rho\sigma} \Big[\langle \overline{u}_\alpha(n) u_\beta(n) \overline{u}_\rho(0) u_\sigma(0) \rangle \\
&\quad + \langle \overline{u}_\alpha(n) u_\beta(n) \overline{u}_\rho(0) u_\sigma(0) \rangle \Big] \\
&= (\gamma_5)_{\alpha\beta} (\gamma_5)_{\rho\sigma} \Big[\langle u_\beta(n) \overline{u}_\alpha(n) \rangle \langle u_\sigma(0) \overline{u}_\rho(0) \rangle \\
&\quad - \langle u_\beta(n) \overline{u}_\rho(0) \rangle \langle u_\sigma(0) \overline{u}_\alpha(n) \rangle \Big] \\
&= (\gamma_5)_{\alpha\beta} (\gamma_5)_{\rho\sigma} \left(D^{-1}_{\beta\alpha}(n|n) D^{-1}_{\sigma\rho}(0|0) - D^{-1}_{\beta\rho}(n|0) D^{-1}_{\sigma\alpha}(0|n) \right).
\end{aligned} \tag{30}$$

We used the anticommutativity of the fermion fields, the Wick's theorem and eq. (8). The other terms can be simplified in a similar manner. The result is

$$\begin{aligned}
\langle \overline{u}(n) \gamma_5 u(n) \overline{d}(0) \gamma_5 d(0) \rangle &= (\gamma_5)_{\alpha\beta} (\gamma_5)_{\rho\sigma} D^{-1}_{\beta\alpha}(n|n) D^{-1}_{\sigma\rho}(0|0), \\
\langle \overline{d}(n) \gamma_5 d(n) \overline{u}(0) \gamma_5 u(0) \rangle &= (\gamma_5)_{\alpha\beta} (\gamma_5)_{\rho\sigma} D^{-1}_{\beta\alpha}(n|n) D^{-1}_{\sigma\rho}(0|0), \\
\langle \overline{d}(n) \gamma_5 d(n) \overline{d}(0) \gamma_5 d(0) \rangle &= (\gamma_5)_{\alpha\beta} (\gamma_5)_{\rho\sigma} \left(D^{-1}_{\beta\alpha}(n|n) D^{-1}_{\sigma\rho}(0|0) - D^{-1}_{\beta\rho}(n|0) D^{-1}_{\sigma\alpha}(0|n) \right).
\end{aligned} \tag{31}$$

Then,

$$\langle P^3(n)P^3(0)\rangle = -\frac{1}{2}(\gamma_5)_{\alpha\beta}D^{-1}_{\beta\rho}(n|0)(\gamma_5)_{\rho\sigma}D^{-1}_{\sigma\alpha}(0|n) = -\frac{1}{2}\text{Tr}\left[\gamma_5 D^{-1}(n|0)\gamma_5 D^{-1}(0|n)\right]. \tag{32}$$

This is the same trace we had in eq. (7) (with a factor of 1/2) and we compute it in the same manner. For the other correlator

$$\langle A_0^3(n)P^3(0)\rangle \tag{33}$$

the procedure is exactly the same. The final result is

$$\langle A_0^3(n)P^3(0)\rangle = -\frac{1}{2}\text{Tr}\left[\gamma_0\gamma_5 D^{-1}(n|0)\gamma_5 D^{-1}(0|n)\right]. \tag{34}$$

We rewrite this expression in a more convenient way which allows us to reuse the propagators for every correlator. Then, we do not have to invert the Dirac matrix many times.

$$\begin{aligned}
-\frac{1}{2}\text{Tr}\left[\gamma_0\gamma_5 D^{-1}(n|0)\gamma_5 D^{-1}(0|n)\right] &= \frac{1}{2}\text{Tr}\left[\gamma_5 D^{-1}(n|0)\gamma_5 D^{-1}(0|n)\gamma_0\right] \\
&= \frac{1}{2}D^{-1}_{\alpha\beta}(n|0)(\gamma_5)_{\beta\rho}D^{-1}_{\rho\sigma}(0|n)(\gamma_5)_{\sigma\lambda}(\gamma_0)_{\lambda\alpha} \\
&= \frac{1}{2}D^{-1}_{\alpha\beta}(n|0)D^{-1*}_{\lambda\beta}(n|0)(\gamma_0)_{\lambda\alpha} \\
&= \frac{1}{2}\left(D^{-1}_{1\beta}(n|0)D^{-1*}_{0\beta}(n|0) + D^{-1}_{0\beta}(n|0)D^{-1*}_{1\beta}(n|0)\right)
\end{aligned} \tag{35}$$

where we used the property $\{\gamma_0, \gamma_5\} = 0$ and eq. (9). As a summary, we have

$$m_{\text{PCAC}}(t) = \frac{C_{\text{AP}}(t+1) - C_{\text{AP}}(t-1)}{4C_{\text{PP}}(t)} \tag{36}$$

$$C_{\text{PP}}(t) = -\frac{1}{2\sqrt{N_x}}\sum_x \sum_{\alpha,\beta=0}^{1} |D^{-1}(t,x|0,0)_{\alpha,\beta}|^2 \tag{37}$$

$$C_{\text{AP}}(t) = \frac{1}{2\sqrt{N_x}}\sum_x \sum_{\beta=0}^{1}\left[D^{-1}_{1\beta}(t,x|0)D^{-1*}_{0\beta}(t,x|0,0) + D^{-1}_{0\beta}(t,x|0,0)D^{-1*}_{1\beta}(t,x|0,0)\right]. \tag{38}$$

3 Results

In Table 1 we show the parameters of distinct simulations and the measurements for m_π and m_{PCAC}. We performed the simulations with 10^3 thermalization sweeps and 10^3 measurements with 10 decorrelation steps. We worked on a 64×64 lattice at $\beta = 4$. All the computations were carried out on a workstation.

In Fig. 1 we show the pion correlator $C_\pi(t)$ in logarithmic scale and the effective mass. To determine m_π we fit the plateau of $m_{\text{eff}}(t)$ in a range far from the middle point, where the effective mass can become unstable close to the critical point. In our case we fit the mass between $t = 10$ and $t = 25$.

In Fig. 2 we show the fermion mass, as measured with eq. (26). The mesaurements of the fermion mass are less susceptible to autocorrelations, as can be seen from the error bars and the small uncertainties in Table 1.

Finally, in Fig. 3 we plot the behaviour of m_π for different values of m_{PCAC}. We compare our results with a prediction by A. Smilga for the two-flavor Schwinger model made in Ref. [2].

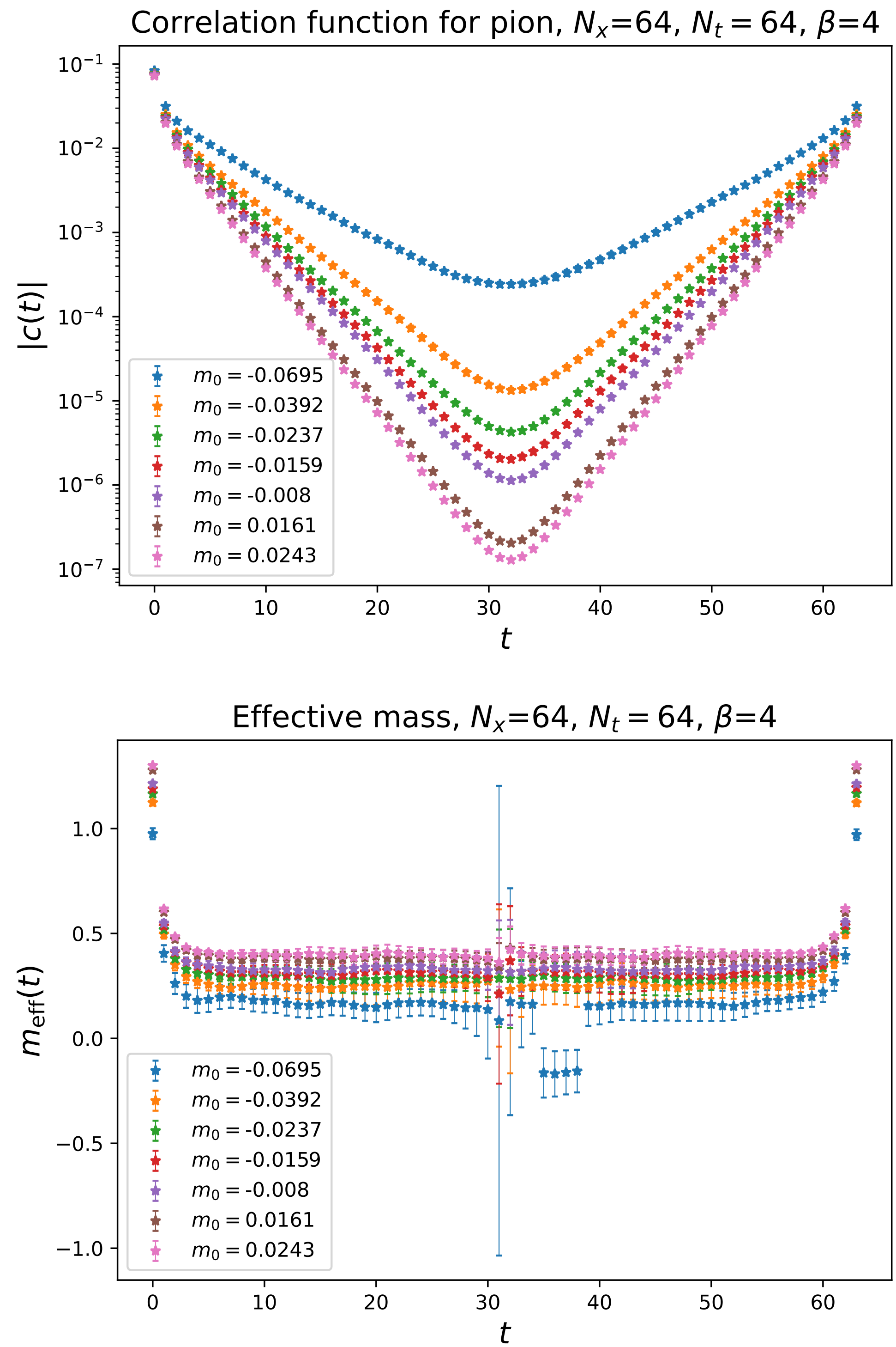

Figure 1: $C_\pi(t)$ and $m_{\text{eff}}(t)$ for different bare masses. The effective mass helps us identifying the plateau, which corresponds to the case when excited states are suppresed and we can extract the ground state. Close to the middle point the correlator becomes unstable. We fit the mass between $t=10$ and $t=25$ in every case.

κ	m_0	Leapfrog steps	Trajectory length	Acceptance rate	MPI Cores	Computing time [s]	m_{PCAC}	m_π
0.259	-0.0695	10	0.25	0.68	16	5839	0.0313(3)	0.1647(162)
0.255	-0.0392	8	0.25	0.80	16	3035	0.0618(4)	0.2488(199)
0.253	-0.0237	10	0.287	0.85	16	3099	0.0767(4)	0.2864(159)
0.252	-0.0159	10	0.287	0.87	16	2843	0.0851(3)	0.3101(141)
0.251	-0.008	10	0.287	0.88	16	2766	0.0933(2)	0.3308(14)
0.248	0.0161	25	1	0.81	16	5054	0.1165(3)	0.3824(83)
0.247	0.0243	25	1	0.81	16	5013	0.1241(3)	0.3981(86)

Table 1: Results and parameters for different simulations at $\beta = 4$, $V = 64^2$. The closer we get to the critical mass $m_{\text{PCAC}} = 0$ the more the simulation should take due to the ill-conditioning of the Dirac matrix. However, the number of leapfrog steps is also very important. In any case, the acceptance rate has to be manually tuned. We performed 10^3 thermalization sweeps, 10^3 measurements and 10 decorrelation steps. Increasing statistics, decorrelation steps and tuning the acceptance rate to be $0.6 \sim 0.7$ can improve the results.

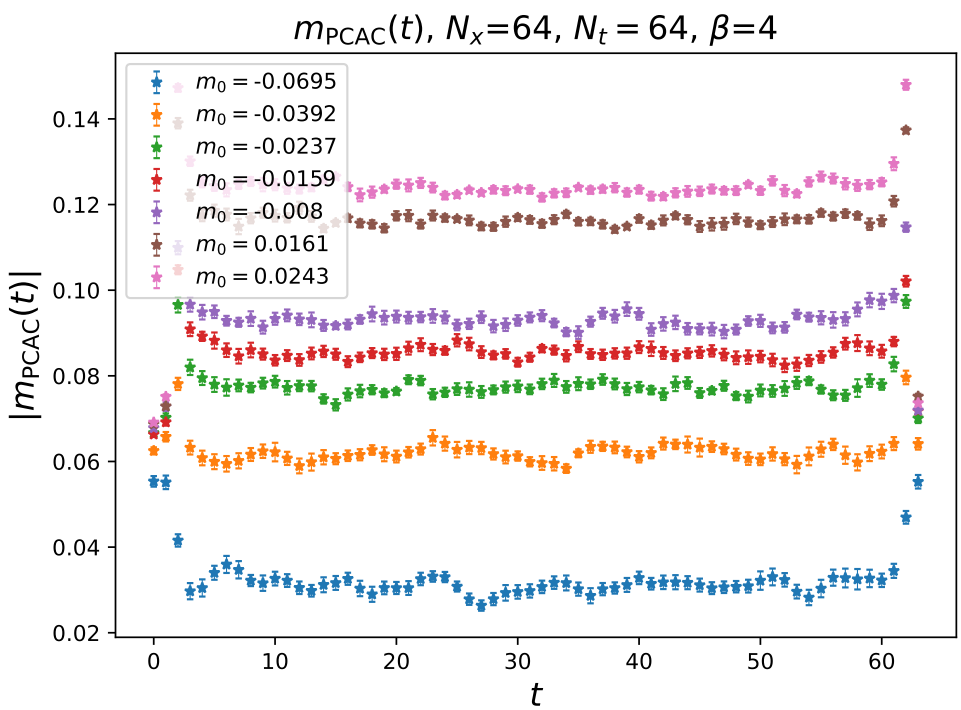

Figure 2: PCAC mass measured with eq. (26) for different values of the bare mass. To fit the plateau we consider the range $t \in [10, 25]$.

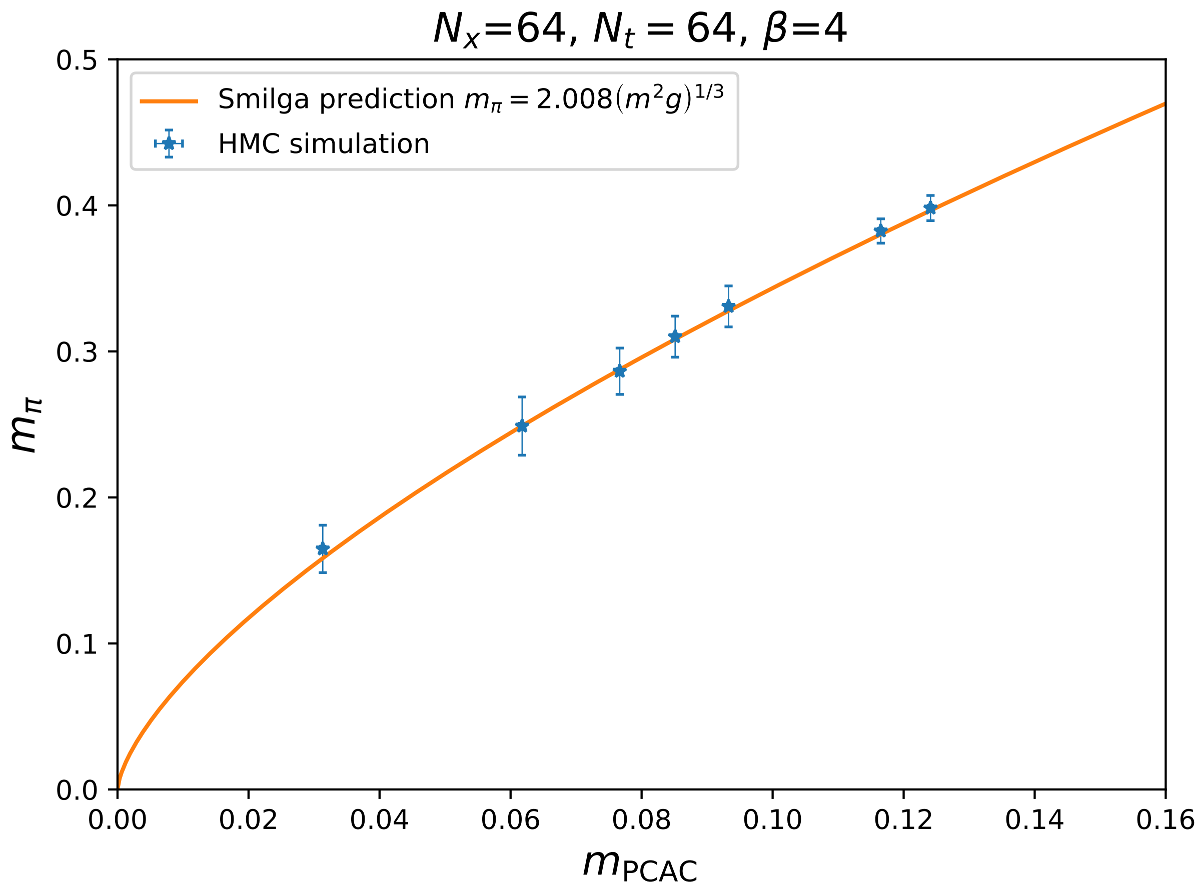

Figure 3: Behaviour of m_π vs. m_{PCAC}. Our results coincide perfectly with the prediction for infinite volume made by Smilga in Ref. [2]. The exact numbers are displayed in Table 1. The gauge couplings is related to $\beta = 1/g^2$.

References

[1] C. Gattringer and C. B. Lang, *Quantum Chromodynamics on the Lattice: An Introductory Presentation*, Lec. Notes Phys. 788 Springer, (2010)

[2] A. V. Smilga, On the fermion condensate in the Schwinger model, Phys. Lett. B **278**, 371 (1992).